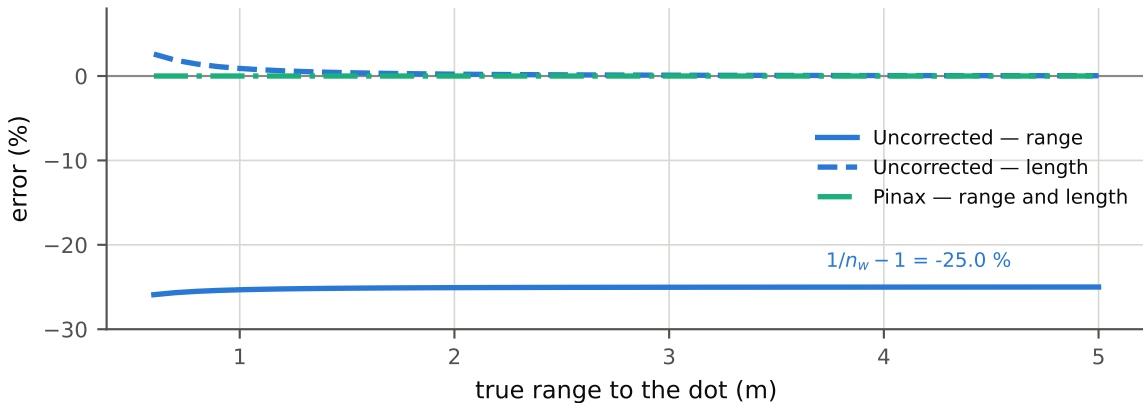


The port's error goes into range, and cancels out of length



A 300 mm fish with the dot on it, laser offset 104 mm. The uncorrected range error is flat because it is $1/n_w$; the length error is what survives the cancellation, and it grows toward close range where the fish subtends a larger angle. Simulation, so the truth is known rather than derived through a camera model.